

OBLIGATORY TRIPLE SYSTEMS AND ERDŐS PROBLEM 593

SAMUIL PETKOV

ABSTRACT. An obligatory triple system is a finite 3-uniform hypergraph that occurs in every 3-uniform hypergraph of uncountable chromatic number. Erdős Problem 593 asks for a characterisation of these systems. We prove that, after isolated vertices are removed, a finite triple system is obligatory if and only if it is linear, every hyperedge-node of its Levi graph is incident with a bridge, and every Berge cycle has even length. Equivalently, the obligatory systems are generated by private-vertex expansions of finite bipartite graphs and finite edgeless systems under disjoint unions and one-point amalgamations. One direction follows from a bridge-block decomposition into bipartite expansion pieces. For the converse, sequence lifts provide uncountably chromatic hosts that exclude each possible failure of the intrinsic conditions. All arguments are carried out in ZFC.

INTRODUCTION

A finite triple system is *obligatory* if it embeds in every triple system of uncountable chromatic number. Erdős asked for a characterisation of all such systems (Erdős, 1995); the question is now catalogued as Erdős Problem 593 (Bloom, 2026). Theorem A gives both a constructive and an intrinsic answer.

The answer has two complementary forms. One is constructive: start with private-vertex expansions of finite bipartite graphs and close under finite disjoint unions and one-point amalgamations. The other is intrinsic and is read from the Levi graph after isolated vertices are removed.

Theorem A.

For every finite triple system F , the following are equivalent:

1. F is obligatory.
2. $F \in \mathcal{B}$.
3. F° is linear, every hyperedge-node of $I(F^\circ)$ is incident with a bridge, and every Berge cycle of F° has even length.

Here F° is obtained by deleting all isolated vertices, and $\mathcal{B}$ is the smallest class containing every private-vertex expansion J^+ of a finite bipartite graph and every finite edgeless triple system, and closed under finite disjoint unions and one-point amalgamations.

The corresponding graph problem was settled by Erdős and Hajnal (1966, Corollary 5.6 and Theorem 7.4): the finite obligatory graphs are precisely the bipartite graphs. The classical linearity obstruction for uniform hypergraphs is due to Erdős et al. (1973, Theorem 2, p. 532). A systematic study of obligatory triple systems was initiated by Erdős et al. (1975) and developed further by Komjáth (2001), Hajnal and Komjáth (2008), and Komjáth (2008). The expansion theorem of Reiher (2024, Theorem 1.2) supplies the positive atoms used below.

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Independent related work. The proof presented here was developed without consulting Li’s manuscript; the comparison in this paragraph was added only after the present argument was complete. Li (2026, Theorems 1.1 and 4.6) states the same classification and proposes an exact bridge-trace theorem for a related sequence lift. The self-contained proof below shows independently that this structural target is correct. No statement or argument from Li’s preprint is used in the proof.

Proof strategy. The positive direction has two parts. Section 3 proves directly that the private-vertex expansion of every finite bipartite graph is obligatory; Section 4 proves closure under disjoint unions and one-point amalgamations. Section 5 identifies the constructive class with the intrinsic Levi-graph conditions by deleting the Levi bridges and reassembling the resulting pieces along a quotient forest.

For the converse, Sections 6–8 construct explicit avoidance hosts. A one-apex sequence lift transfers uncountable chromatic number from a graph to a triple system while rigidly controlling every finite linear trace. Nonlinearity is excluded by an Erdős–Rado host, failure of the bridge condition by a sequence lift of K_{ω_1} , and an odd Berge cycle by a sequence lift of an uncountably chromatic graph of sufficiently large odd girth. This proves all three implications needed for Theorem A.

Set-theoretic scope. All arguments are in ZFC. The only non-elementary external results used as black boxes are the de Bruijn–Erdős compactness theorem (de Bruijn and Erdős, 1951, Theorem 1, pp. 371–373) and the pair case of the Erdős–Rado partition theorem (Erdős and Rado, 1956, Theorem 4(i), formula (95), p. 471),

$$(2^{\aleph_0})^+ \longrightarrow (\aleph_1)_{\aleph_0}^2.$$

All embeddings are non-induced.

1. PRELIMINARIES AND DEFINITIONS

A *triple system* is a simple 3-uniform hypergraph $H = (V(H), E(H))$, so $E(H) \subseteq [V(H)]^3$. A colouring $c : V(H) \rightarrow \lambda$ is *proper* when no hyperedge is monochromatic, and $\chi(H)$ is the least cardinal λ admitting such a colouring. An embedding of a triple system F in H is an injective map $f : V(F) \rightarrow V(H)$ such that $f[e] \in E(H)$ for every $e \in E(F)$; embeddings are not required to be induced. A finite triple system F is *obligatory* if every triple system H with $\chi(H) > \aleph_0$ contains an embedding of F . A point is *isolated* if it belongs to no hyperedge, and F° denotes the result of deleting all isolated points.

All ordinary graphs in this paper are simple. A triple system is *linear* when any two distinct hyperedges meet in at most one point. Its *Levi graph* $I(F)$ is the bipartite graph with classes $V(F)$ and $E(F)$, where a point-node p is adjacent to a hyperedge-node e exactly when $p \in e$. A *bridge* is an edge of a graph whose deletion increases the number of connected components. A *Berge cycle of length* $\ell \geq 2$ consists of distinct points $p_0, \dots, p_{\ell-1}$ and distinct hyperedges $e_0, \dots, e_{\ell-1}$ such that $p_i \in e_i \cap e_{i+1}$, with indices modulo ℓ . Equivalently, it is a simple cycle of length 2ℓ in $I(F)$.

For a finite simple graph J , its *private-vertex expansion* J^+ has vertex set

$$V(J) \dot{\cup} \{p_a : a \in E(J)\}$$

and hyperedges $\{u, v, p_{\{u,v\}}\}$ for $\{u, v\} \in E(J)$, where the points p_a are new and pairwise distinct. A *one-point amalgamation* of vertex-disjoint triple systems F_0, F_1 is obtained by choosing $x_i \in V(F_i)$, identifying x_0 with x_1 , and making no other identifications or new hyperedges. Finite disjoint unions and finite edgeless systems include the empty cases.

FINITE REDUCTIONS

Lemma 1.1: isolated-vertex reduction.

Let F be a finite triple system and let F° be obtained by deleting all isolated vertices. Then

$$F \text{ is obligatory} \iff F^\circ \text{ is obligatory},$$

and

$$F \in \mathcal{B} \iff F^\circ \in \mathcal{B}.$$

Proof. If F° is obligatory and H has uncountable chromatic number, then H has infinitely many vertices. After embedding the finite system F° , map the isolated vertices of F injectively to unused host vertices. Containment is non-induced, so additional host edges are irrelevant. The converse follows because F° is a subhypergraph of F .

For membership in $\mathcal{B}$, the implication $F^\circ \in \mathcal{B} \Rightarrow F \in \mathcal{B}$ follows by taking the disjoint union with a finite edgeless system.

For the converse, use structural induction on a construction of F in $\mathcal{B}$. If $F = J^+$, delete the isolated graph vertices of J to obtain J_0 ; then $(J^+)^\circ = J_0^+ \in \mathcal{B}$. If F is edgeless, its reduction is the empty edgeless system. Reduction commutes with disjoint union.

Suppose finally that F is a one-point amalgamation of F_0 and F_1 , identifying $x_i \in V(F_i)$. By induction $F_0^\circ, F_1^\circ \in \mathcal{B}$. If both x_i are nonisolated in their factors, then F° is their one-point amalgamation at those vertices. If exactly one x_i is nonisolated, the isolated side disappears at the identified point and F° is the disjoint union of the two reductions. The same disjoint-union description applies when both x_i are isolated, because the identified point is deleted. Hence $F^\circ \in \mathcal{B}$ in every case. $\square$

Thus isolated vertices may be suppressed throughout the substantive argument and restored at the end.

Lemma 1.2: finite deletion.

If $\chi(H) > \aleph_0$ and $S \subseteq V(H)$ is finite, then

$$\chi(H - S) > \aleph_0.$$

Proof. Otherwise a countable colouring of $H - S$, together with finitely many fresh colours for S , would colour H countably. $\square$

COLOURING AND CLOSURE TOOLS

Lemma 1.3: two elementary colouring facts.

The following facts will be used repeatedly.

1. If $E(H) = E_1 \cup \dots \cup E_r$ and every $(V(H), E_i)$ is countably chromatic, then H is countably chromatic.
2. If a graph G admits an orientation in which every vertex has outdegree at most $d < \omega$, then $\chi(G) \leq 2d + 1$.

Proof. For the first assertion, take the product of the r countable colourings.

For the second, every finite subgraph has at most $d|V|$ edges and therefore a vertex of degree at most $2d$. It is $(2d + 1)$ -colourable by greedy deletion. The de Bruijn–Erdős compactness theorem (de Bruijn and Erdős, 1951, Theorem 1) then gives a $(2d + 1)$ -colouring of all of G . $\square$

Lemma 1.4: closure-chain lemma.

Let κ be an uncountable cardinal, let $1 \leq r < \omega$, and let

$$\Phi : [\kappa]^r \longrightarrow [\kappa]^{<\omega}$$

have uniformly finite values. There is an increasing continuous chain

$$\emptyset = M_0 \subseteq M_1 \subseteq \cdots \subseteq M_i \subseteq \cdots \quad (i < \text{cf } \kappa)$$

such that

$$\bigcup_{i < \text{cf } \kappa} M_i = \kappa, \quad |M_i| < \kappa,$$

and every M_i is Φ -closed:

$$a \in [M_i]^r \implies \Phi(a) \subseteq M_i.$$

Proof. Choose sets $A_i \subseteq \kappa$, each of cardinality below κ , whose union is κ . At a successor stage close $M_i \cup A_i$ under Φ , iterating the operation ω times. Since Φ is finitary and finite-valued, this does not increase an infinite cardinal below κ . At a limit stage take the union. A union of an increasing chain of sets closed under a finitary operation remains closed; and because the stage is below $\text{cf } \kappa$, its cardinality remains below κ . $\square$

Write

$$I_i = M_{i+1} \setminus M_i.$$

Then $(I_i)_{i < \text{cf } \kappa}$ partitions κ , and $M_i = \bigcup_{j < i} I_j$.

2. THE GRAPH LEMMA NEEDED FOR THE EXPANSION ATOM

The next lemma is the finite complete-bipartite consequence of Erdős and Hajnal (1966, Corollary 5.6, p. 72). We include its short closure-chain proof because the same construction is used later.

Lemma 2.1: uncountable chromatic number forces complete bipartite graphs.

For every positive integer n , every graph of uncountable chromatic number contains $K_{n,n}$.

Proof. Suppose otherwise, and choose a $K_{n,n}$ -free graph G of uncountable chromatic number whose vertex cardinality κ is minimal.

For $A \in [\kappa]^n$, let

$$N(A) = \{v \in \kappa : v \text{ is adjacent to every member of } A\}.$$

Because G is $K_{n,n}$ -free, $|N(A)| < n$. Apply Lemma 1.4 to $A \mapsto N(A)$, and let M_i, I_i be the resulting chain and layers.

By minimality of κ , every induced graph $G[I_i]$ is countably chromatic. Let $G^\times$ consist of the edges joining distinct layers, and orient each such edge from the later endpoint toward the earlier endpoint. If $v \in I_i$ had n earlier neighbours, those neighbours would form an n -set $A \subseteq M_i$, and closure would give

$$v \in N(A) \subseteq M_i,$$

contrary to $v \in I_i$. Thus the orientation of $G^\times$ has outdegree at most $n - 1$. By Lemma 1.3, $G^\times$ is finitely colourable.

Colour each $G[I_i]$ with colours in ω , reusing the same palette on all layers, and combine this with a finite proper colouring of $G^\times$. The resulting countable product colouring is proper on all of G , a contradiction. $\square$

3. PRIVATE-VERTEX EXPANSIONS OF BIPARTITE GRAPHS ARE OBLIGATORY

RAINBOW BIPARTITE SUBMATRICES

Lemma 3.1: rainbow bipartite submatrices.

For all positive integers n, t , there exists $q = q(n, t)$ with the following property. Suppose the edges of $K_{q,q}$ are coloured, with an arbitrary colour set, so that at every vertex each individual colour occurs on fewer than t incident edges. Then there are n -element subsets X' and Y' of the two vertex classes such that all n^2 edges between X' and Y' have distinct colours.

Proof. Choose independent uniformly random n -subsets X' and Y' . For every unordered pair P of distinct same-coloured edges, let I_P be the indicator that both edges of P lie in the selected $K_{n,n}$, and put

$$Z = \sum_P I_P.$$

Thus $Z = 0$ exactly when all selected edges have distinct colours.

Let m_γ be the number of edges of colour γ . The local bound gives

$$m_\gamma \leq (t-1)q.$$

At a fixed vertex and for a fixed colour γ , fewer than t incident edges have colour γ . Summing $\binom{d_{v,\gamma}}{2} \leq (d_{v,\gamma})(t-1)/2$ over the $2q$ vertices shows that the number of same-coloured pairs sharing a vertex is at most $(t-1)q^2$. The total number of same-coloured pairs is at most

$$\sum_\gamma \binom{m_\gamma}{2} \leq \frac{1}{2} (\max_\gamma m_\gamma) \sum_\gamma m_\gamma \leq \frac{1}{2} (t-1)q^3.$$

If the two edges share a vertex in the left class, their exact selection probability is

$$\frac{n}{q} \frac{\binom{n}{2}}{\binom{q}{2}};$$

the same formula holds when the shared vertex is on the right. If the two edges are disjoint, the exact probability is

$$\left(\frac{\binom{n}{2}}{\binom{q}{2}} \right)^2.$$

For $q \geq 2n$, these are at most $2n^3/q^3$ and $4n^4/q^4$, respectively. By linearity of expectation,

$$\mathbb{E}[Z] \leq \frac{2(t-1)n^3}{q} + \frac{2(t-1)n^4}{q}.$$

This is below 1 for sufficiently large q . Since Z is integer-valued and nonnegative, it cannot satisfy $Z \geq 1$ for every choice when $\mathbb{E}[Z] < 1$. Hence some choice has $Z = 0$, and the selected $K_{n,n}$ is rainbow. $\square$

THE COMPLETE BIPARTITE EXPANSION ATOM

Proposition 3.2: the complete bipartite expansion atom.

For every positive integer n , the triple system $K_{n,n}^+$ is obligatory.

Proof. Suppose not. Choose a $K_{n,n}^+$ -free triple system

$$H = (\kappa, E)$$

with $\chi(H) > \aleph_0$ and with κ minimal.

For a pair xy , write

$$d_H(xy) = |\{z : \{x, y, z\} \in E\}|.$$

Set $t = 3n^2 + 1$ and define a graph R on κ by

$$xy \in E(R) \iff d_H(xy) \geq t.$$

Claim 3.2.1: R is countably chromatic. Otherwise Lemma 2.1 gives a $K_{n,n}$ in R , with core vertex set C , $|C| = 2n$. For each of its n^2 graph edges xy , choose a third vertex p_{xy} with

$$\{x, y, p_{xy}\} \in E,$$

avoiding C and all previously chosen private vertices. At every stage fewer than $2n + n^2 < t$ vertices are forbidden, while xy has at least t possible third vertices. Thus the p_{xy} can be chosen distinct and outside C , producing a copy of $K_{n,n}^+$, a contradiction. $\square$

Let

$$E_1 = \{e \in E : \text{some pair contained in } e \text{ is an edge of } R\}, \quad E_2 = E \setminus E_1.$$

A proper colouring of R is a proper colouring of (κ, E_1) , since every E_1 -edge contains an R -edge. Hence $\chi(\kappa, E_1) \leq \aleph_0$, and therefore

$$\chi(\kappa, E_2) > \aleph_0. \tag{3.1}$$

For a pair xy , define

$$\Phi(xy) = \begin{cases} \{z : \{x, y, z\} \in E\}, & d_H(xy) < t, \\ \emptyset, & d_H(xy) \geq t. \end{cases}$$

Apply Lemma 1.4 to Φ , obtaining M_i, I_i .

Every $e \in E_2$ has at least two vertices in its highest layer. Indeed, suppose the highest layer met e only in $z \in I_i$. Then the other pair $xy = e \setminus \{z\}$ lies in M_i . Since $e \in E_2$, $xy \notin R$, so $d_H(xy) < t$. Closure gives $z \in \Phi(xy) \subseteq M_i$, a contradiction.

Thus every E_2 -edge is either contained in one layer, or has exactly two vertices in its highest layer I_i and its third vertex in M_i . Let E_i^* be the E_2 -edges contained in I_i . Since $|I_i| < \kappa$ and H is $K_{n,n}^+$ -free, minimality of κ gives $\chi(I_i, E_i^*) \leq \aleph_0$. Consequently

$$E^* = \bigcup_i E_i^*$$

is countably chromatic. By (3.1), the remaining crossing-edge system

$$E^{**} = E_2 \setminus E^*$$

has uncountable chromatic number.

For each i , define a graph G_i on I_i by declaring $xy \in E(G_i)$ when there is a $\beta \in M_i$ such that $\{x, y, \beta\} \in E^{**}$. If every G_i were countably chromatic, colour each I_i properly in G_i , using the same countable palette on every layer. Every edge of E^{**} would then have differently coloured top-layer vertices, giving a countable colouring of (κ, E^{**}) , a contradiction. Hence some G_i is uncountably chromatic.

Choose $q = q(n, t)$ from Lemma 3.1. Lemma 2.1 gives a $K_{q,q}$ in G_i , with sides X, Y . For every $xy \in X \times Y$, choose

$$\beta_{xy} \in M_i$$

such that $\{x, y, \beta_{xy}\} \in E^{**}$, and regard β_{xy} as the colour of the edge xy .

This edge-colouring is locally $(t - 1)$ -bounded. For example, if for fixed x and fixed β there were t different vertices y with $\beta_{xy} = \beta$, then the pair $\{x, \beta\}$ would have codegree at least t . It would belong to R , forcing the triples $\{x, y, \beta\}$ into E_1 , contrary to their membership in $E^{**} \subseteq E_2$. The same argument applies at vertices of Y .

Lemma 3.1 gives n -sets $X' \subseteq X, Y' \subseteq Y$ for which the $\beta_{xy}, x \in X', y \in Y'$, are all distinct. They lie in M_i , whereas $X' \cup Y' \subseteq I_i$, so none is a core vertex. The triples

$$\{x, y, \beta_{xy}\}, \quad x \in X', y \in Y',$$

form $K_{n,n}^+$, the final contradiction. $\square$

Corollary 3.3: every finite bipartite expansion is obligatory.

If J is any finite bipartite graph, then J^+ is obligatory.

Proof. Embed the two vertex classes of J in the two classes of some $K_{n,n}$. Then J^+ is a subhypergraph of $K_{n,n}^+$. Obligatoriness is downward closed under non-induced containment: a copy of the larger finite system contains a copy of the smaller one. Apply Proposition 3.2. $\square$

Corollary 3.3 is also a consequence of Reiher (2024, Theorem 1.2); the preceding argument gives a direct proof in the present notation.

4. CLOSURE OF OBLIGATORINESS UNDER THE OPERATIONS DEFINING $\mathcal{B}$

Closure under disjoint unions and one-point amalgamations is part of the established theory; see Komjáth (2001, pp. 233–238) and the summary in Reiher (2024, p. 1). We include complete proofs because the rooted abundance argument is used below.

DISJOINT-UNION CLOSURE

Lemma 4.1: disjoint-union closure.

The class of finite obligatory triple systems is closed under finite disjoint unions.

Proof. Let F_1, F_2 be obligatory and let H be uncountably chromatic. First find a copy of F_1 . Delete its finite vertex set. By Lemma 1.2 the remaining triple system is still uncountably chromatic, so it contains F_2 . The two copies are disjoint. Iteration proves the finite case. $\square$

ROOTED ABUNDANCE AND ONE-POINT AMALGAMATION

Definition 4.2: rooted abundance.

Let (F, r) be a finite rooted triple system with $|V(F)| > 1$. For a triple system H and $m \geq 1$, let $R_m(F, r; H)$ be the set of vertices v for which there are m rooted copies of F , all sending r to v , whose off-root vertex sets are pairwise disjoint.

Lemma 4.3: rooted-abundance lemma.

If (F, r) is finite, $|V(F)| > 1$, and F is obligatory, and if $\chi(H) > \aleph_0$, then

$$\chi(H[R_m(F, r; H)]) > \aleph_0.$$

Proof. Put

$$B = V(H) \setminus R_m(F, r; H).$$

For each $v \in B$, consider the family of off-root vertex sets of all copies of F rooted at v . Because $v \notin R_m(F, r; H)$, this family has matching number at most $m - 1$. Choose a maximal pairwise disjoint subfamily. It contains at most $m - 1$ members; let S_v be their union. Then

$$|S_v| \leq (m - 1)(|V(F)| - 1) =: D,$$

Moreover, S_v meets the off-root part of every rooted copy at v : otherwise that off-root set would be disjoint from every member of the chosen subfamily and could be added to it, contradicting maximality.

Form a graph D_B on B by joining v to each member of $S_v \cap B$. If an edge is generated in both directions, assign it to one of its generators, and orient it away from that generator. Every vertex has outdegree at most D , so Lemma 1.3 gives a finite proper colouring of D_B .

Let A be one colour class. Then

$$S_v \cap A = \emptyset \quad (v \in A).$$

Consequently $H[A]$ contains no copy of F . Indeed, if φ were such a copy and $v = \varphi(r)$, its off-root part would have to meet S_v although both are subsets of A . Since F is obligatory, every F -free triple system is countably chromatic. Thus each of the finitely many colour classes of D_B induces a countably chromatic subsystem. Tagging a vertex by its finite D_B -colour and its countable colour inside that class gives a countable proper colouring of $H[B]$.

If $H[R_m(F, r; H)]$ were countably chromatic as well, tagging the two sides would give a countable colouring of H . This contradicts $\chi(H) > \aleph_0$. $\square$

Proposition 4.4: one-point amalgamation closure.

The class of finite obligatory triple systems is closed under one-point amalgamations.

Proof. Let F_1, F_2 be obligatory, with selected vertices r_1, r_2 , and let F be their one-point amalgamation. If one factor consists only of its selected vertex, the assertion is immediate. Otherwise set

$$m = |V(F_2)|.$$

By Lemma 4.3, the set

$$R = R_m(F_1, r_1; H)$$

induces an uncountably chromatic subsystem of every uncountably chromatic H . Hence $H[R]$ contains a copy of F_2 ; write φ for its embedding and let $v = \varphi(r_2)$.

At v there are m rooted copies of F_1 with pairwise disjoint off-root sets. The set $\varphi(V(F_2) \setminus \{r_2\})$ has $m - 1$ vertices, so it meets at most $m - 1$ of those off-root sets. One rooted F_1 -copy therefore avoids the F_2 -copy outside v . Their union is the required one-point amalgamation. $\square$

Corollary 3.3, Lemma 4.1, Proposition 4.4, and the trivial edgeless case prove

$$(2) \implies (1). \tag{4.1}$$

5. EQUIVALENCE OF THE CONSTRUCTIVE AND INTRINSIC DESCRIPTIONS

We prove $(2) \iff (3)$. Isolated vertices may be ignored by Lemma 1.1.

FORWARD PRESERVATION

Proposition 5.1: the intrinsic conditions are preserved by the generators and operations.

Every generator of $\mathcal{B}$ satisfies the intrinsic conditions, and the conditions are preserved under finite disjoint unions and one-point amalgamations.

Proof. For a bipartite graph J , the expansion J^+ is linear. Every hyperedge-node is joined to its private point by a bridge. A Berge cycle cannot use a private point, so the Berge cycles of J^+ are precisely the ordinary graph cycles of J , with the same lengths; these are even. Edgeless systems satisfy the conditions vacuously, and disjoint union plainly preserves them.

Consider a one-point amalgamation. Hyperedges belonging to different factors intersect only in the identified point, so linearity is preserved. Its Levi graph is the vertex-sum of the two Levi graphs at a point-node. Every simple cycle in a vertex-sum lies wholly in one factor: a cycle crossing into the other factor would have to pass through the cut vertex twice. Consequently bridges in the factors remain bridges, and every Berge cycle lies in one factor and retains its parity. $\square$

Thus

$$(2) \implies (3). \tag{5.1}$$

BRIDGE-BLOCK RECONSTRUCTION**Proposition 5.2: bridge-block decomposition.**

Suppose F is finite, has no isolated vertices, and satisfies the three intrinsic conditions. Then $F \in \mathcal{B}$.

Proof. If $E(F) = \emptyset$, then the assumption that F has no isolated vertices gives $V(F) = \emptyset$, and F is an edgeless generator. Hence assume $E(F) \neq \emptyset$. Put $L = I(F)$, let $B(L)$ be its set of bridges, and let $\mathcal{C}$ be the set of connected components of $L - B(L)$.

Consider a hyperedge-node e . Since e is incident with a bridge and has degree three, it has at most two nonbridge incidences. It cannot have exactly one: a nonbridge incidence lies on a cycle, and that cycle must leave e through a second nonbridge incidence. Therefore every hyperedge-node has either no nonbridge incidences or exactly two.

The nontrivial components. Let $C \in \mathcal{C}$ contain a nonbridge incidence. Every hyperedge-node $e \in C$ has exactly two point-neighbours in C . Contract each such e to an ordinary graph edge between those two point-neighbours. This produces a finite graph J_C .

The graph J_C is simple: parallel edges would correspond to two hyperedges of F sharing two points, contrary to linearity. Every cycle in J_C gives a Berge cycle of the same length in F . Hence J_C has no odd cycle and is bipartite.

Each hyperedge-node $e \in C$ has a third point-neighbour p_e across a bridge. These third points satisfy:

1. $p_e \notin C$; otherwise a path in C from p_e to e , together with the incidence ep_e , would place that incidence on a cycle.
2. If $e \neq f$ lie in C , then $p_e \neq p_f$; otherwise a path in C from e to f , together with ep_e and fp_f , would place those bridge incidences on a cycle.

Define a map from J_C^+ to the subsystem P_C formed by these hyperedges as follows. Send every graph vertex of J_C to the corresponding point-node of C , send the private vertex indexed by the contracted edge associated with e to p_e , and send that expanded edge to the hyperedge e of F . The two observations above show that the vertex map is injective: the private points are pairwise distinct and lie outside C . Every e has exactly the two contracted endpoints and p_e as its three vertices, so this map is incidence-preserving and surjective on the vertices and hyperedges of P_C . Hence it is an isomorphism $J_C^+ \cong P_C$.

Components consisting of one hyperedge-node. If a hyperedge-node e has no nonbridge incidence, it is an isolated component after the bridges are deleted. Its single triple is the expansion of a one-edge bipartite graph. Thus every hyperedge of F belongs to a piece P_C isomorphic to J^+ for a finite bipartite graph J . The hyperedge sets of these pieces partition $E(F)$. Their vertex sets cover $V(F)$: every point is incident with a hyperedge, and it either lies in that hyperedge-node's bridge-deleted component or is the point endpoint of a bridge leaving it.

Quotient forest. Contract every component of $L - B(L)$. The original bridges become edges of a quotient multigraph T . Two distinct bridges cannot join the same pair of bridge-deleted components: paths inside those two components, together with the two bridges, would form a cycle containing both, contradicting that they are bridges. Thus T is a simple graph.

To see that T is a forest, suppose that it contained a cycle. Within each contracted component on this cycle, choose a path joining the endpoints of the two incident quotient edges. Concatenating these paths with the corresponding bridges gives a closed walk in L that traverses each of those bridges exactly once. This is impossible: deleting a bridge separates its endpoints, so every closed walk crosses the resulting cut an even number of times. Hence T is a forest.

A vertex $C \in V(T) = \mathcal{C}$ is called *active* if the bridge-deleted component C contains a hyperedge-node. For an active C , the piece P_C consists of the hyperedges represented in C , all point-nodes of C , and the point endpoints of bridge incidences leaving hyperedge-nodes in C . An inactive component contains no hyperedge-node and hence no incidence of $L - B(L)$, so it is a singleton point-node.

Claim 5.2.1: running intersection.

Every component of T contains an active vertex, because F has no isolated vertices. Choose an active root in each component, and order the active vertices by nondecreasing depth, with arbitrary order among equal-depth vertices. When an active component C is added, its piece P_C meets the union of the earlier pieces in at most one point.

Proof of the claim. For a point $p \in V(F)$, let $X(p) \in \mathcal{C}$ be the component of $L - B(L)$ containing the point-node p . The active pieces containing p are precisely $P_{X(p)}$, when $X(p)$ is active, together with the pieces P_D at active neighbours D of $X(p)$ whose corresponding original bridge has point endpoint p . Thus the vertices of T indexing pieces that contain p are contained in the closed star centred at $X(p)$.

Suppose that $p \in P_C$ also lies in an earlier piece. If $X(p) = C$, every other piece containing p is indexed by a neighbour of C , and only the parent neighbour can be earlier. If $X(p) \neq C$, then C is a neighbour of $X(p)$. The vertex $X(p)$ cannot be a child of C , because then $X(p)$ and all its other neighbours in the star would be deeper than C . Hence $X(p)$ is the parent of C . In either case, p is the point endpoint of the unique parent edge of C . The same reasoning applies to every point shared with an earlier piece, so all such points coincide with this parent point p_C , and

$$P_C \cap \bigcup_{D \text{ earlier}} P_D \subseteq \{p_C\}.$$

For a root active component the intersection is empty. $\square$

Starting with the root piece in each component, add the active pieces in this order. If the new intersection is empty, use a disjoint union. If it is the singleton $\{p_C\}$, use a one-point amalgamation at that point. Several earlier pieces may already contain the same inactive attachment point, but the new step still identifies only that one existing vertex.

For completeness, the identity maps from the pieces into F assemble inductively to an incidence isomorphism from the constructed system onto the union of the pieces already added. Indeed, the running-intersection claim shows that the next piece and the preceding union have either no common vertex or exactly the one vertex used in the amalgamation; hence the construction makes no unintended

identification. The piece hyperedge sets are disjoint and partition $E(F)$, while their vertex sets cover $V(F)$. After all active pieces have been added, the resulting subsystem is therefore all of F . Different components of T are combined by disjoint union. Hence $F \in \mathcal{B}$. $\square$

This proves

$$(3) \implies (2). \quad (5.2)$$

Combining (5.1) and (5.2),

$$(2) \iff (3). \quad (5.3)$$

6. THE SEQUENCE LIFT USED IN THE AVOIDANCE CONSTRUCTIONS

THE ONE-APEX CONSTRUCTION

Let $G = (X, A)$ be an arbitrary graph, where $A = E(G)$ is also regarded as an alphabet. Let

$$T = A^{<\omega_1}$$

be the tree of all sequences of graph edges of countable ordinal length. For $s \in T$ and $x \in X$, write $x_s = (s, x)$.

Definition 6.1: the one-apex lift.

Define the triple system $\mathcal{L}(G)$ by

$$V(\mathcal{L}(G)) = T \times X.$$

For $s \in T$, an edge $a = \{x, y\} \in A$, a sequence $t \in T$ extending $s \frown a$, and $z \in X$, put

$$\{x_s, y_s, z_t\} \in E(\mathcal{L}(G)). \quad (6.1)$$

The two vertices x_s, y_s form the base pair and z_t is the apex. Since t properly extends s , these are three distinct vertices. In the resulting triple, the sequence node occurring twice uniquely identifies s ; its two graph labels identify the edge $a = \{x, y\}$, and the remaining vertex identifies t, z . Thus each hyperedge arises from unique data, up to interchanging x and y , and the system is simple and 3-uniform.

CHROMATIC AND FINITE-TRACE PROPERTIES

Lemma 6.2: chromatic lower bound.

If $\chi(G) > \aleph_0$, then

$$\chi(\mathcal{L}(G)) > \aleph_0.$$

Proof. Suppose $c : T \times X \rightarrow \omega$ is proper. For every $s \in T$, the restriction $x \mapsto c(x_s)$ is a countable colouring of G . Hence there is an edge

$$a_s = \{x_s^0, x_s^1\} \in E(G)$$

whose two vertices at node s have a common colour k_s .

Construct $s_\alpha \in A^\alpha$ for $\alpha < \omega_1$ by

$$s_0 = \emptyset, \quad s_{\alpha+1} = s_\alpha \frown a_{s_\alpha},$$

and at limit ordinals take unions.

If $\alpha < \beta$, then s_β extends $s_\alpha \widehat{a}_{s_\alpha}$. If $k_{s_\alpha} = k_{s_\beta}$, take either endpoint z of the monochromatic edge at s_β . By (6.1),

$$\{(s_\alpha, x_{s_\alpha}^0), (s_\alpha, x_{s_\alpha}^1), (s_\beta, z)\}$$

is a monochromatic hyperedge. Therefore the colours k_{s_α} , $\alpha < \omega_1$, are pairwise distinct, impossible in ω . $\square$

Theorem 6.3: exact finite linear trace theorem.

Every finite linear subhypergraph K of $\mathcal{L}(G)$ is obtainable, by finite disjoint unions and one-point amalgamations, from systems J^+ where J is a finite subgraph of G .

Proof. First set aside every isolated vertex of K . Each is a one-vertex edgeless factor, hence isomorphic to J^+ for a one-vertex edgeless subgraph J of G , and may be restored by disjoint union at the end. We may therefore assume that every vertex of K lies in a hyperedge.

Every hyperedge of $\mathcal{L}(G)$ has a unique base node s : exactly two of its vertices have sequence coordinate s , while the apex has a strictly longer sequence coordinate. Group the hyperedges of K by their base nodes, and let K_s be the subhypergraph whose hyperedges are those based at s and whose vertices are their union.

For a fixed graph edge $a = \{x, y\} \in E(G)$, at most one hyperedge of K_s can have base pair $\{x_s, y_s\}$, since two such hyperedges would share two vertices and violate linearity. Let J_s be the finite subgraph of G whose edges are precisely the graph edges used by hyperedges of K_s , with vertex set consisting of their endpoints.

Define a map $J_s^+ \rightarrow K_s$ by sending each graph vertex $x \in V(J_s)$ to the core vertex x_s , and by sending the private vertex of $a = \{x, y\}$ to the apex of the unique hyperedge of K_s with base pair $\{x_s, y_s\}$. Distinct graph edges have distinct apices: if z_t were the apex for a and a' , the coordinate of t immediately after s would have to equal both a and a' . No apex is a core vertex, because an apex has sequence coordinate strictly extending s , whereas every core vertex has coordinate s . The map is therefore injective, and by construction it is bijective on hyperedges and preserves each incidence. Hence $K_s \cong J_s^+$.

Now consider intersections between different K_s . If a point z_t lies in both K_s and K_u , then both s and u are prefixes of t , so they are comparable. Suppose $s \subsetneq u$. A common point cannot be a core point of K_s ; it is an apex of K_s . The graph edge used by its hyperedge in K_s is the coordinate of u immediately following s . Hence all points in $K_s \cap K_u$ would be apices of the same base edge of K_s . There is at most one such apex. Therefore

$$|V(K_s) \cap V(K_u)| \leq 1. \quad (6.2)$$

Construct a bipartite incidence graph Q whose one class is the set of active base nodes s , and whose other class consists of points lying in at least two of the K_s ; join s to p when $p \in K_s$.

We claim that Q is a forest. Suppose it has a cycle, and choose on that cycle a base node s of minimum length. Its two neighbouring base nodes on the cycle are proper descendants of s . The two intervening shared points are distinct, so they are distinct apices in K_s , arising from two distinct graph edges $a \neq a'$. Thus the two neighbouring base nodes lie in different immediate branches below s , one beginning with a and the other with a' .

Starting from one neighbour of s , every base node on the remaining part of the cycle lies in the same immediate branch below s . Indeed, consecutive base nodes share a point and are therefore comparable, and two comparable proper descendants of s cannot lie in different immediate branches. Induction along the remaining path therefore keeps every base node in the first branch, contradicting that its final node lies in the other branch. Hence Q is acyclic.

Root each component of Q at a base node, and order its base nodes by nondecreasing even distance from the root. For a nonroot base node s , let p_s be the point-node adjacent to s on the path to the root.

Claim 6.3.1: base-fibre running intersection.

Every earlier factor that meets K_s meets it at p_s .

Proof of the claim. Suppose K_u is earlier and contains a point $p \in K_s$. Then Q contains the length-two path $s - p - u$. Let the distance of s from the root be $2d$. The point-node p has distance either $2d - 1$ or $2d + 1$. In the second case u has distance $2d + 2$, so it cannot be earlier. Thus p has distance $2d - 1$, and the uniqueness of the path to the root in the tree Q gives $p = p_s$. This also covers an earlier sibling u , which shares the same parent point p_s . $\square$

Equation (6.2) and the claim give

$$V(K_s) \cap \bigcup_{u \text{ earlier}} V(K_u) \subseteq \{p_s\}.$$

Starting with the root factor, add the remaining factors in this order by disjoint unions or one-point amalgamations as appropriate. At each step the natural maps into K identify exactly the displayed common point and nothing else, so they assemble to an isomorphism onto the union of the factors already added. Different components of Q , and the isolated factors removed at the start, are combined by disjoint union. $\square$

Corollary 6.4: restrictions on finite linear traces.

Every finite linear $K \subseteq \mathcal{L}(G)$ satisfies:

1. *every hyperedge-node of $I(K)$ is incident with a bridge;*
2. *every Berge cycle of K is contained in a single factor J_s^+ ;*
3. *a Berge cycle of length ℓ in K yields an ordinary cycle of length ℓ in G .*

Proof. In every J_s^+ the incidence with the private vertex is a bridge. Bridgehood is preserved by one-point amalgamation. A simple Levi cycle cannot cross a one-point cut, so every Berge cycle lies in one factor. Berge cycles in J_s^+ are exactly ordinary graph cycles in J_s . $\square$

7. UNCOUNTABLY CHROMATIC GRAPHS OF LARGE ODD GIRTH

The shift-graph mechanism used here is classical; compare Erdős and Hajnal (1966, Section 7, especially Theorem 7.4). We include proofs of the precise cardinal and odd-girth bounds needed below.

CHROMATIC GROWTH OF SHIFT GRAPHS

For a positive integer r and an infinite cardinal κ , define the directed shift graph $D_r(\kappa)$ as follows:

- its vertices are increasing r -tuples

$$(\alpha_0, \dots, \alpha_{r-1}), \quad \alpha_0 < \dots < \alpha_{r-1} < \kappa;$$

- it has an arc

$$(\alpha_0, \dots, \alpha_{r-1}) \longrightarrow (\alpha_1, \dots, \alpha_r)$$

whenever $\alpha_0 < \dots < \alpha_r$.

Let $S_r(\kappa)$ be its underlying graph. Write

$$\beth_0(\aleph_0) = \aleph_0, \quad \beth_{j+1}(\aleph_0) = 2^{\beth_j(\aleph_0)}.$$

Lemma 7.1: chromatic number of shift graphs.

If

$$\kappa > \beth_{r-1}(\aleph_0),$$

then

$$\chi(S_r(\kappa)) > \aleph_0.$$

Proof. The case $r = 1$ is immediate because $S_1(\kappa) = K_\kappa$. Assume $r \geq 2$. Suppose $S_r(\kappa)$ has a proper colouring c with λ colours. For every increasing $(r-1)$ -tuple u , define

$$C(u) = \{c(u \frown \beta) : \beta > \max u\} \subseteq \lambda.$$

If

$$u = (\alpha_0, \dots, \alpha_{r-2}), \quad v = (\alpha_1, \dots, \alpha_{r-1})$$

are adjacent in $S_{r-1}(\kappa)$, then

$$c(\alpha_0, \dots, \alpha_{r-1}) \in C(u).$$

It cannot belong to $C(v)$, since that would give some $\beta > \alpha_{r-1}$ for which the adjacent r -tuples

$$(\alpha_0, \dots, \alpha_{r-1}), \quad (\alpha_1, \dots, \alpha_{r-1}, \beta)$$

have the same colour. Hence $C(u) \neq C(v)$. Thus a λ -colouring of $S_r(\kappa)$ yields a 2^λ -colouring of $S_{r-1}(\kappa)$.

Starting with a countable colouring of $S_r(\kappa)$ and iterating $r-1$ times would give a colouring of

$$S_1(\kappa) = K_\kappa$$

with at most $\beth_{r-1}(\aleph_0)$ colours, impossible when κ is larger. $\square$

ODD-GIRTH GROWTH

Lemma 7.2: odd girth of shift graphs.

The odd girth of $S_r(\kappa)$ is at least

$$2r + 1.$$

Proof. For a digraph D , let δD be its arc digraph: the vertices of δD are the arcs of D , and

$$(u, v) \longrightarrow (v, w)$$

is an arc of δD . We have

$$D_{r+1}(\kappa) \cong \delta D_r(\kappa).$$

We use the following fact: if D is acyclic and the underlying graph of D has odd girth g , then the underlying graph of δD has odd girth at least $g + 2$. Indeed, a directed cycle in δD would concatenate to a directed closed walk in D , which would contain a directed cycle; hence δD is also acyclic.

Take an odd cycle $a_0, \dots, a_{m-1}$ in the underlying graph of δD , with indices modulo m . For the adjacency of a_{i-1} and a_i , put $\varepsilon_i = +$ when $\text{head}(a_{i-1}) = \text{tail}(a_i)$, and put $\varepsilon_i = -$ in the reverse case. Since δD is acyclic, the cyclic sign sequence is not constant. Its number c of sign changes is therefore a positive even integer, so $c \geq 2$.

Let v_i be the common endpoint in D of the consecutive arcs a_{i-1} and a_i . If $\varepsilon_i = \varepsilon_{i+1}$, then v_i and v_{i+1} are the two endpoints of a_i , in one order or the other. If $\varepsilon_i \neq \varepsilon_{i+1}$, then $v_i = v_{i+1}$: both are the tail

of a_i in the $+$, $-$ case and both are its head in the $-$, $+$ case. Thus $v_0, \dots, v_{m-1}, v_0$ is a closed walk in the underlying graph of D with exactly c stationary steps. Deleting those steps gives a closed walk of length $m - c$. Since m is odd and c is even, $c \neq m$. Together with $c \geq 2$, this gives

$$1 \leq m - c \leq m - 2,$$

and $m - c$ is odd.

Every odd closed walk contains an odd cycle. Indeed, if a vertex occurs twice internally, split the walk at two consecutive occurrences; one of the two resulting closed walks has odd length. Repeating this reduction produces an odd closed walk with no repeated internal vertex, hence an odd cycle. Therefore $m - c \geq g$, and so $m \geq g + 2$.

Now $D_1(\kappa)$ is the transitive tournament; it is acyclic and its underlying graph K_κ has odd girth 3. Iterating the preceding observation gives odd girth at least

$$3 + 2(r - 1) = 2r + 1.$$

□

Corollary 7.3: ZFC graphs with uncountable chromatic number and prescribed odd girth.

For every finite m there is a graph G with $\chi(G) > \aleph_0$ and no odd cycle of length at most m .

Proof. Choose r with $2r + 1 > m$, and let

$$G = S_r(\boxminus_{r-1}(\aleph_0)^+).$$

Apply Lemmas 7.1 and 7.2. □

8. THE AVOIDANCE HOSTS

We prove

$$\neg(3) \implies \neg(1).$$

NONLINEARITY OBSTRUCTION

Proposition 8.1: avoidance of every nonlinear finite triple system.

There is an uncountably chromatic linear triple system. Consequently, every finite nonlinear triple system is non-obligatory.

Proof. We use the Erdős–Rado relation (Erdős and Rado, 1956, Theorem 4(i), formula (95), p. 471)

$$(2^{\aleph_0})^+ \longrightarrow (\aleph_1)_{\aleph_0}^2. \quad (8.1)$$

Let $\kappa = (2^{\aleph_0})^+$. Define a triple system T_κ whose vertices are the pairs in $[\kappa]^2$, and whose hyperedges are

$$\{\{\alpha, \beta\}, \{\alpha, \gamma\}, \{\beta, \gamma\}\}, \quad \alpha < \beta < \gamma < \kappa.$$

This triple system is linear: two distinct triangles of a complete graph cannot share two graph edges, because two graph edges of a triangle determine the third edge and the three underlying vertices.

It is uncountably chromatic. A countable vertex-colouring of T_κ is a countable edge-colouring of K_κ . By (8.1) there is a monochromatic subset of cardinality $\aleph_1$, and in particular a monochromatic graph triangle. This is a monochromatic hyperedge of T_κ .

If a finite triple system F is nonlinear, it has two distinct edges sharing at least two vertices. No injective embedding of those edges can exist in the linear host T_κ . Thus T_κ is F -free. □

MISSING-BRIDGE OBSTRUCTION

Proposition 8.2: avoidance when the bridge condition fails.

Let F be finite and linear, and suppose some hyperedge-node of $I(F^\circ)$ is incident with no bridge. Then F is non-obligatory.

Proof. By Lemma 1.1 it is enough to avoid F° , so assume that F has no isolated vertices. Take $G = K_{\omega_1}$. Then $\chi(G) > \aleph_0$, and Lemma 6.2 gives

$$\chi(\mathcal{L}(G)) > \aleph_0.$$

If F appeared in $\mathcal{L}(G)$, retain exactly the host hyperedges corresponding to the edges of F . Because the embedding is injective and F is linear, these selected host hyperedges form a finite linear subhypergraph isomorphic to F . Corollary 6.4 says that every hyperedge-node of such a trace is incident with a bridge, contradicting the chosen hyperedge of F . Therefore $\mathcal{L}(K_{\omega_1})$ is an uncountably chromatic F -free host. $\square$

ODD-CYCLE OBSTRUCTION

Proposition 8.3: avoidance of an odd Berge cycle.

Let F be finite and linear, and suppose F° has an odd Berge cycle. Then F is non-obligatory.

Proof. Again suppress isolated vertices. Put $m = |E(F)|$. Choose r with $2r + 1 > m$, and set

$$G = S_r(\boxminus_{r-1}(\aleph_0)^+).$$

By Lemmas 7.1 and 7.2, $\chi(G) > \aleph_0$, and G has no odd cycle of length at most m . Lemma 6.2 gives

$$\chi(\mathcal{L}(G)) > \aleph_0.$$

If F appeared in $\mathcal{L}(G)$, the selected image hyperedges would form a finite linear trace. By Corollary 6.4, every Berge cycle of that trace lies in one factor J_s^+ and induces an ordinary graph cycle of the same length in $J_s \subseteq G$.

The odd Berge cycle of F has length at most m , so G would contain an odd cycle of length at most m , a contradiction. Thus $\mathcal{L}(G)$ is an uncountably chromatic F -free host. $\square$

The three propositions cover every failure of the intrinsic conditions, and prove

$$\neg(3) \implies \neg(1). \tag{8.2}$$

9. COMPLETION OF THE PROOF

Sections 3 and 4 prove $(2) \implies (1)$. Section 5 proves $(2) \iff (3)$, and Section 8 proves the contrapositive $\neg(3) \implies \neg(1)$. These implications establish all equivalences in Theorem A. In particular,

$$F \text{ is obligatory} \iff F \in \mathcal{B},$$

$$F^\circ \text{ is linear,}$$

$$F \in \mathcal{B} \iff \begin{array}{l} \text{every hyperedge-node of } I(F^\circ) \text{ is incident with a bridge,} \\ \text{every Berge cycle of } F^\circ \text{ has even length.} \end{array}$$

All embeddings are non-induced: additional host hyperedges on the image vertices are irrelevant throughout. The proof uses only ZFC, the displayed successor cardinals, graph-colouring compactness, and the standard Erdős–Rado relation (8.1). It assumes neither CH nor GCH, and no forcing axiom or large-cardinal hypothesis.

FORMALISATION

A partial Lean 4 formalisation developed using ChatGPT-5.6 and Aristotle (Achim et al., 2025) is available in a self-contained source file on GitHub.

AI ASSISTANCE

ChatGPT-5.6 Sol Pro and Aristotle were consulted for proof development, adversarial checking, and the Lean formalisation. In particular, ChatGPT-5.6 Sol Pro proposed the central idea of the converse in Sections 6–8: the one-apex sequence lift $\mathcal{L}(G)$ and its exact finite-linear-trace decomposition. The author reviewed all incorporated suggestions and assumes full responsibility for the manuscript.

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